HARD · ALGEBRA

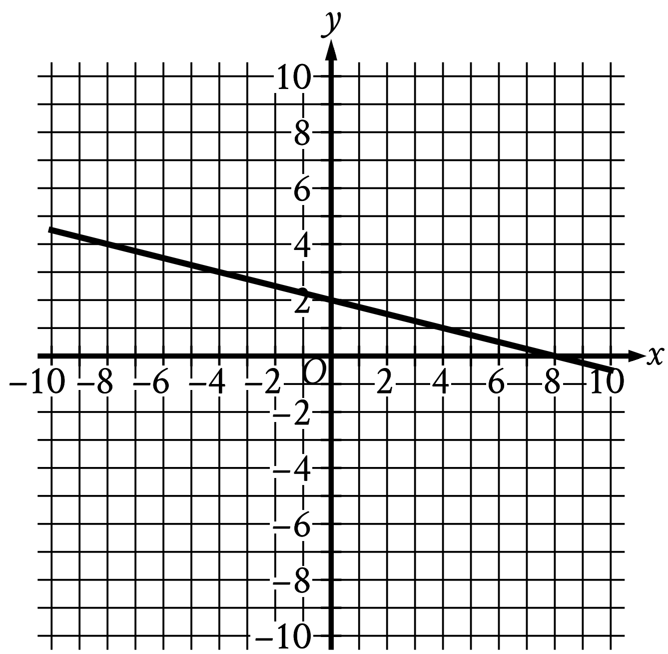
Algebra

09

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29



- The line slants gradually down from left to right.
- The line passes through the following points:
  - (negative 8 comma 4)
  - (0 comma 2)
  - (8 comma 0)

The graph of $y = fx + 14$ is shown. Which equation defines function f?

A. $fx = -\frac{1}{4}x - 12$

B. $fx = -\frac{1}{4}x + 16$

C. $fx = -\frac{1}{4}x + 2$

D. $fx = -\frac{1}{4}x - 14$

Q2 ALGEBRA

The cost of renting a backhoe for up to 10 days is \$ 270 for the first day and \$ 135 for each additional day. Which of the following equations gives the cost y, in dollars, of renting the backhoe for x days, where x is a positive integer and $x \leq 10$?

A. $y = 270x - 135$

B. $y = 270x + 135$

C. $y = 135x + 270$

D. $y = 135x + 135$

$$y < 6x + 2$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	20
5	32
7	44

B.

x	y
3	16
5	36
7	40

C.

x	y
3	16
5	28
7	40

D.

x	y
3	24
5	36
7	48

Q4

ALGEBRA

How many solutions does the equation $1015x - 9 = -156 - 10x$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Q5 ALGEBRA

Which system of linear equations has no solution?

A. $-2x + 3y = -9$
$2x - 3y = 9$

B. $2x - 3y = 9$
$3x + 4y = 10$

C. $2x - 3y = 9$
$-6x + 9y = -27$

D. $-2x + 3y = 9$
$4x - 6y = 18$

Q6 ALGEBRA

$$\frac{3}{5}x + \frac{3}{4}y = 7$$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
1	$\frac{113}{20}$
2	$\frac{101}{20}$
4	$\frac{77}{20}$

B.

x	y
1	$\frac{47}{5}$
2	$\frac{44}{5}$
4	$\frac{38}{5}$

C.

x	y
1	$\frac{148}{15}$
2	$\frac{136}{15}$
4	$\frac{112}{15}$

D.

x	y
1	$\frac{128}{15}$
2	$\frac{116}{15}$

4	$\frac{92}{15}$

Q7

GRID-IN ALGEBRA

The equation $h = \frac{9v - 273.15}{5} + 32$ gives the corresponding temperature h, in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. If a substance has a temperature of 467.33 degrees Fahrenheit, what is the corresponding temperature, in kelvins, of this substance?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8 ALGEBRA

A business owner plans to purchase the same model of chair for each of the 81 employees. The total budget to spend on these chairs is $\$ 14{,}000$, which includes a 7% sales tax. Which of the following is closest to the maximum possible price per chair, before sales tax, the business owner could pay based on this budget?

A. $\$ 148.15$

B. $\$ 161.53$

C. $\$ 172.84$

D. $\$ 184.94$

Q9

GRID-IN ALGEBRA

$$\frac{12x + 28}{4} - \frac{s}{13} = rx - 8$$

In the given equation, s and r are constants, and $s > 0$. If the equation has infinitely many solutions, what is the value of s?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

ALGEBRA

$$2x + 9y = 7$$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

I. $3x + 13.5y = 10.5$
II. $3x - 13.5y = 10.5$

A. I only

B. II only

C. I and II

D. Neither I nor II